Terms (implementation flow)

Laplacian (7 points) $\nabla_h^2 u_i = (\sum_{j \in N(i)} u_j - 6u_i)/h^2$ ($N(i)$: the 6 neighbours), second order; discretises $\beta \nabla^2 \phi_{3D}$

Mirror (walls) a ghost cell outside the wall with $u_{\text{ghost}} = u_{\text{in}}$: zero flux $\partial u/\partial n = 0$ at second order

Upwind the face value of the advection $\mathbf{v} \cdot \nabla u$ is taken from the upstream side; first order is monotone but adds diffusion $\approx |\mathbf{v}|h/2$

MUSCL, minmod, TVD MUSCL: a linear reconstruction in each cell gives second-order face values. minmod(a, b): the smaller one if the signs agree, else 0 (zero slope at extrema). The total variation $\sum_i |u_{i+1} - u_i|$ does not grow (TVD: no new extrema)

Regularised front direction $\mathbf{n}_c = \nabla c / \sqrt{|\nabla c|^2 + \varepsilon^2}$, $\varepsilon^2 = 10^{-6}$: a unit vector for $|\nabla c| \gg \varepsilon$, continuously 0 for $|\nabla c| \to 0$

RK4 explicit four-stage Runge-Kutta, local error $O(\Delta t^5)$; stable below limits on $\beta \Delta t/h^2$ and the Courant number $|\mathbf{v}|\Delta t/h$

Backward Euler (Eq. 35) $\dot{c}$ replaced by $(c^{n+1} - c^n)/\Delta t$, right-hand side at the new time: stable for any Δt, one linear solve per step.
Diffusion time $\ell^2/d \approx 10^{-8} T^* \ll \Delta t$, so the time term is $\sim 10^{-6}$; the limit without it differs by $\leq 0.1\,\%$ (32^3)

One-step lag step 2 uses $(\Sigma_i \bar{\phi}_i)^n$ instead of $(\Sigma_i \bar{\phi}_i)^{n+1}$ (explicit coupling), error $O(\Delta t)$; two iterations change less than the step error

Sparse matrix, CG, AMG $-\nabla_h^2 + q$ is symmetric positive definite with 7 non-zeros per row. CG: Krylov iteration for such systems.
AMG: a preconditioner built from coarse levels of the matrix; the iteration count hardly grows with the grid

Implicit Euler $(\mathbf{g}^{k+1} - \mathbf{g}^k)/\delta t = \mathbf{f}(\mathbf{g}^{k+1})$: first order, A-stable

Newton, Jacobian $\mathbf{Q}(\mathbf{g}) = \mathbf{0}$ solved by $\mathbf{g} \leftarrow \mathbf{g} - \mathbf{J}^{-1}\mathbf{Q}$, $\mathbf{J} = \partial \mathbf{Q}/\partial \mathbf{g}$; quadratic convergence near the solution

ctypes Python's foreign-function interface: calls the compiled Fortran library on double arrays (the same routine as in ANSYS)

Gauss points, state variables the $2 \times 2 \times 2$ Gauss-Legendre points of an element; state variables are kept per point between increments (ustatev)

NEM, MLS a second-order Taylor expansion (9 unknowns) fitted to the values of 30 neighbouring Gauss points by least squares with Gaussian weights (width $\beta^* = 0.2$ mm); its coefficients give ∇u and $\nabla^2 u$ (moving least squares)

Neo-Hookean $W = C_{10}(\bar{I}_1 - 3) + (J-1)^2/D_1$: isochoric and volumetric parts, $C_{10} = \mu/2$ (Klempt et al. 2024, Eq. 20)

Selective reduced integration ($\bar{\mathbf{B}}$) the volumetric strain is averaged over the element, avoiding volumetric locking for $\nu \to 0.5$

Perturbation of F F_{ij} shifted by 10^{-7} (relative); forward differences of the stress give the tangent $\partial \boldsymbol{\sigma}/\partial \boldsymbol{\varepsilon}$ (Sun et al. 2008)